

## SEMESTER S6

### SYSTEMS LAB

|                                            |                 |                    |               |
|--------------------------------------------|-----------------|--------------------|---------------|
| <b>Course Code</b>                         | <b>PCCSL607</b> | <b>CIE Marks</b>   | 50            |
| <b>Teaching Hours/Week<br/>(L: T:P: R)</b> | 0:0:3:0         | <b>ESE Marks</b>   | 50            |
| <b>Credits</b>                             | 2               | <b>Exam Hours</b>  | 2 Hrs 30 Min. |
| <b>Prerequisites (if any)</b>              | None            | <b>Course Type</b> | Lab           |

#### Course Objectives:

1. To introduce the fundamental concepts of compiler design, including lexical analysis, syntax analysis, and code generation.
2. To equip students with practical skills to design and implement the components of a compiler using tools like LEX and YACC.
3. To teach students the basic and advanced techniques of virtual machine instantiation and management using open-source hypervisors / public cloud platforms.

| <b>Expt.<br/>No.</b> | <b>Experiments</b>                                                                                                                                                                                                   |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b>             | Design and implement a lexical analyzer using C language to recognize all valid tokens in the input program. The lexical analyzer should ignore redundant spaces, tabs and newlines. It should also ignore comments. |
| <b>2</b>             | Write a lex program to display the number of lines, words and characters in an input text.                                                                                                                           |
| <b>3</b>             | Generate a YACC specification to recognize a valid arithmetic expression that uses operators +, − , *, / and parenthesis.                                                                                            |
| <b>4</b>             | Implementation of Calculator using LEX and YACC                                                                                                                                                                      |
| <b>5</b>             | Convert the BNF rules into YACC form and write code to generate abstract syntax tree.                                                                                                                                |
| <b>6</b>             | Write a program to find First and Follow of any given grammar.                                                                                                                                                       |
| <b>7</b>             | Design and implement a recursive descent parser for a given grammar.                                                                                                                                                 |
| <b>8</b>             | Construct a Shift Reduce Parser for a given language.                                                                                                                                                                |
| <b>9</b>             | Write a program to perform constant propagation.                                                                                                                                                                     |

|           |                                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10</b> | Implement Intermediate code generation for simple expressions.                                                                                                                                                                                                |
| <b>11</b> | Implement the back end of the compiler which takes the three address code and produces assembly language instructions that can be assembled and run using a corresponding assembler. The target assembly instructions can be simple move, add, sub, jump etc. |
| <b>12</b> | Instantiation of VMs with image file using open-source hypervisors / public cloud platforms.                                                                                                                                                                  |
| <b>13</b> | Virtual machine Cluster set up using open-source hypervisors / public cloud platforms.                                                                                                                                                                        |
| <b>14</b> | Setting host name for virtual machine nodes in cluster and ssh set up for remote login.                                                                                                                                                                       |
| <b>15</b> | Copy a file from one virtual machine to another virtual machine.                                                                                                                                                                                              |

**Course Assessment Method**  
(CIE: 50 marks, ESE: 50 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Preparation/Pre-Lab Work<br/>experiments, Viva and Timely<br/>completion of Lab Reports / Record<br/>(Continuous Assessment)</b> | <b>Internal<br/>Examination</b> | <b>Total</b> |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------|
| <b>5</b>          | <b>25</b>                                                                                                                           | <b>20</b>                       | <b>50</b>    |

**End Semester Examination Marks (ESE):**

| <b>Procedure/<br/>Preparatory<br/>work/Design/<br/>Algorithm</b> | <b>Conduct of experiment/<br/>Execution of work/<br/>troubleshooting/<br/>Programming</b> | <b>Result with<br/>valid inference/<br/>Quality of<br/>Output</b> | <b>Viva<br/>voce</b> | <b>Record</b> | <b>Total</b> |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------|---------------|--------------|
| <b>10</b>                                                        | <b>15</b>                                                                                 | <b>10</b>                                                         | <b>10</b>            | <b>5</b>      | <b>50</b>    |

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                             | Bloom's Knowledge Level (KL) |
|----------------|-----------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Implement lexical and syntax analyzer using the tools LEX and YACC          | <b>K3</b>                    |
| <b>CO2</b>     | Develop Top-Down and Bottom-Up parsers.                                     | <b>K3</b>                    |
| <b>CO3</b>     | Implement intermediate code for expressions.                                | <b>K3</b>                    |
| <b>CO4</b>     | Experiment with a cluster of virtual machines in a virtualized environment. | <b>K3</b>                    |
| <b>CO5</b>     | Demonstrate the data sharing and communication between virtual machines.    | <b>K3</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 2   | 2   | 3   | 3   | 3   |     |     |     |     | 2    |      | 3    |
| <b>CO2</b> | 2   | 2   | 3   | 3   | 3   |     |     |     | 2   | 2    |      | 3    |
| <b>CO3</b> | 2   | 2   | 3   | 3   | 3   |     |     |     | 2   | 2    |      | 3    |
| <b>CO4</b> |     |     | 3   | 3   | 3   |     |     |     | 2   | 2    |      | 3    |
| <b>CO5</b> |     |     |     |     | 3   |     |     |     | 2   | 2    |      | 3    |

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| <b>Text Books</b> |                                                                                     |                                           |                              |                         |
|-------------------|-------------------------------------------------------------------------------------|-------------------------------------------|------------------------------|-------------------------|
| <b>Sl. No</b>     | <b>Title of the Book</b>                                                            | <b>Name of the Author/s</b>               | <b>Name of the Publisher</b> | <b>Edition and Year</b> |
| 1                 | Engineering a Compiler                                                              | Keith D. Cooper Linda Torczon             | Katey Birtcher               | 3/e, 2023               |
| 2                 | Lex and Yacc                                                                        | John R Levine, Tony Mason& Doug Brown     | O'Reilly Media, Inc          | 2/e, 2013               |
| 3                 | Distributed and Cloud Computing: From Parallel Processing to the Internet of Things | Kai Hwang, Jack Dongarra, Geoffrey C. Fox | Morgan Kaufman               | 1/e, 2018               |
| 4                 | Virtual Machines                                                                    | Manan Shah, Charusmita Shah               | Lambert Academic Publishing  | 1/e, 2018               |

| <b>Reference Books</b> |                                               |                                 |                                |                         |
|------------------------|-----------------------------------------------|---------------------------------|--------------------------------|-------------------------|
| <b>Sl. No</b>          | <b>Title of the Book</b>                      | <b>Name of the Author/s</b>     | <b>Name of the Publisher</b>   | <b>Edition and Year</b> |
| 1                      | Compilers Principles Techniques and Tools     | Aho A Ravi Sethi and J D Ullman | Addison Wesley                 | 2/e, 2013               |
| 2                      | Compiler Construction Principles and Practice | Kenneth C Louden                | Cenage Learning Indian Edition | 1/e, 2007               |
| 3                      | System programming and operating system       | D M Dhamdhare                   | Tata McGraw Hill & Company     | 1/e, 2013               |
| 4                      | The Theory and Practice of Compiler Writing   | Tremblay and Sorenson           | Tata McGraw Hill & Company     | 1/e, 1985               |

| <b>Video Links (NPTEL, SWAYAM...)</b> |                                                                                                                             |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>No.</b>                            | <b>Link ID</b>                                                                                                              |
| 1                                     | <a href="https://archive.nptel.ac.in/courses/106/105/106105190/">https://archive.nptel.ac.in/courses/106/105/106105190/</a> |
| 2                                     | <a href="https://www.virtualbox.org/">https://www.virtualbox.org/</a>                                                       |

## **Continuous Assessment (25 Marks)**

### **1. Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

### **2. Conduct of Experiments (7 Marks)**

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

### **3. Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

### **4. Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

***Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.***

## **Evaluation Pattern for End Semester Examination (50 Marks)**

### **1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.

- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

## **2. Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

## **3. Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

## **4. Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

## **5. Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

# **SEMESTER 7**

**COMPUTER SCIENCE AND  
ENGINEERING**